

Microprocessor and Computer Architecture

BCA — Semester 2 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is microprocessor? Draw a well labelled block diagram of 8085 microprocessor and explain its different components.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are different instructions available in SAP 1 computer? Explain the fetch and execution cycle of LDA instruction of SAP 1 computer along with its timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is micro-program and micro-instruction? Describe symbolic micro-program for instruction FETCH routine of basic computer.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Write a program to perform 8 bit division of data stored in memory location 8050 by data stored in memory location 8051 and store the quotient in memory location 8052 and remainder in 8053 memory location.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain different Logical Microoperations with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain the register organization of basic computer.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain memory hierarchy present in the computer system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is DMA? Explain the role of DMA controller in data transfer.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is pipelining? Explain the four segment instruction pipelining.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Perform 2's complement multiplication of (15) x (-13) using Booth's Multiplication algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is addressing mode? Explain different addressing mode in 8085 microprocessor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a) RISC b) Page Fault

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.